TIDE: Learning to Memorize at Test Time with Surprise-Gated Updates

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Abstract

Attention is an excellent short-term memory but cannot scale to millions of tokens. We introduce TIDE, a neural long-term memory module that keeps learning while the model is being used. The memory is a small multilayer perceptron whose weights are updated at test time by gradient descent on an associative loss, so that it memorizes the tokens that surprise it most. TIDE reaches 71.3% accuracy on LongRecall at 16K tokens and scales to inputs of two million tokens.

1 Introduction

Models with a fixed state, such as linear attention, compress the past into a matrix and lose detail as the context grows. Models with attention keep every token but pay for it in memory. Test-time training offers a third option: treat memorization as learning, and let a parametric memory adapt to each input as it streams by.

We take inspiration from human memory, where events that violate expectations are remembered better. In TIDE, surprise is the gradient of the associative loss with respect to the memory parameters: a token whose value the memory predicts badly produces a large gradient and changes the memory a lot, while an expected token leaves it almost untouched.

2 The Memory Module

The memory M is a two-layer perceptron that maps keys to values. Each incoming token is projected to a key k_t and a value v_t, and the memory is trained online to minimise the reconstruction error of the pair:

$$\text{loss}(M; x_t) = \| M(k_t) - v_t \|^2 \tag{1}$$

A single gradient step per token would react only to the present token. Instead, momentum carries the surprise of earlier tokens, so that the tokens following a surprising event are also memorized even when each of them looks unremarkable on its own. The update with momentum eta_t and step size theta_t is:

$$U_t = \eta_t U_{t-1} - \theta_t \nabla \text{loss}(M_{t-1}; x_t) \tag{2}$$

$$M_t = (1 - a_t) M_{t-1} + U_t \tag{3}$$

Old information is removed by a data-dependent forgetting gate a_t in Equation (3). When the input changes topic the gate approaches one and clears most of the memory; within a topic it stays near zero. All three coefficients eta_t, theta_t and a_t are predicted from the token by small linear layers.

3 Adding Memory to a Model

We compare memory as context, memory as gate and memory as layer. As context, the memory's output for the current segment is prepended to the attention window. As a gate, the memory and a sliding-window attention run in parallel and their outputs are mixed. As a layer, the memory is stacked in front of the attention blocks.

Training uses a parallel form: the sequence is split into segments of 512 tokens, the gradients inside a segment are computed with respect to the memory at the segment start, and momentum is accumulated with a parallel scan.

4 Experiments

We evaluate on the synthetic LongRecall benchmark, which hides 64 facts in filler text and asks for one of them, and on language modelling with the WebMix corpus. All models have 760M parameters.

Table 3: LongRecall accuracy (%) by context length.

Model	4K	16K	64K	256K
Transformer (window 4K)	92.1	24.0	6.2	1.5
Linear attention	58.4	31.7	12.9	4.1
TIDE as context	93.0	71.3	60.8	48.2
TIDE as layer	90.6	66.5	55.1	41.0

Memory as context is the strongest variant. Removing momentum lowers accuracy at 16K tokens from 71.3% to 52.9%, and removing the forgetting gate makes training unstable beyond 64K tokens.

5 Conclusion

Treating memory as a model that learns at test time gives long-context recall that neither attention nor fixed-state recurrences reach at this scale. The cost is a slower forward pass: TIDE is 1.4 times slower than a Transformer of the same size at 4K tokens.

References

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